

What is this?

This notebook is part of a collection of `pluto` notebooks on various topics discussed during the Time Domain Astrophysics course delivered by Stefano Covino at the Università dell'Insubria in Como (Italy). Please direct questions and suggestions to stefano.covino@inaf.it.

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TIME-DOMAIN ASTROPHYSICS

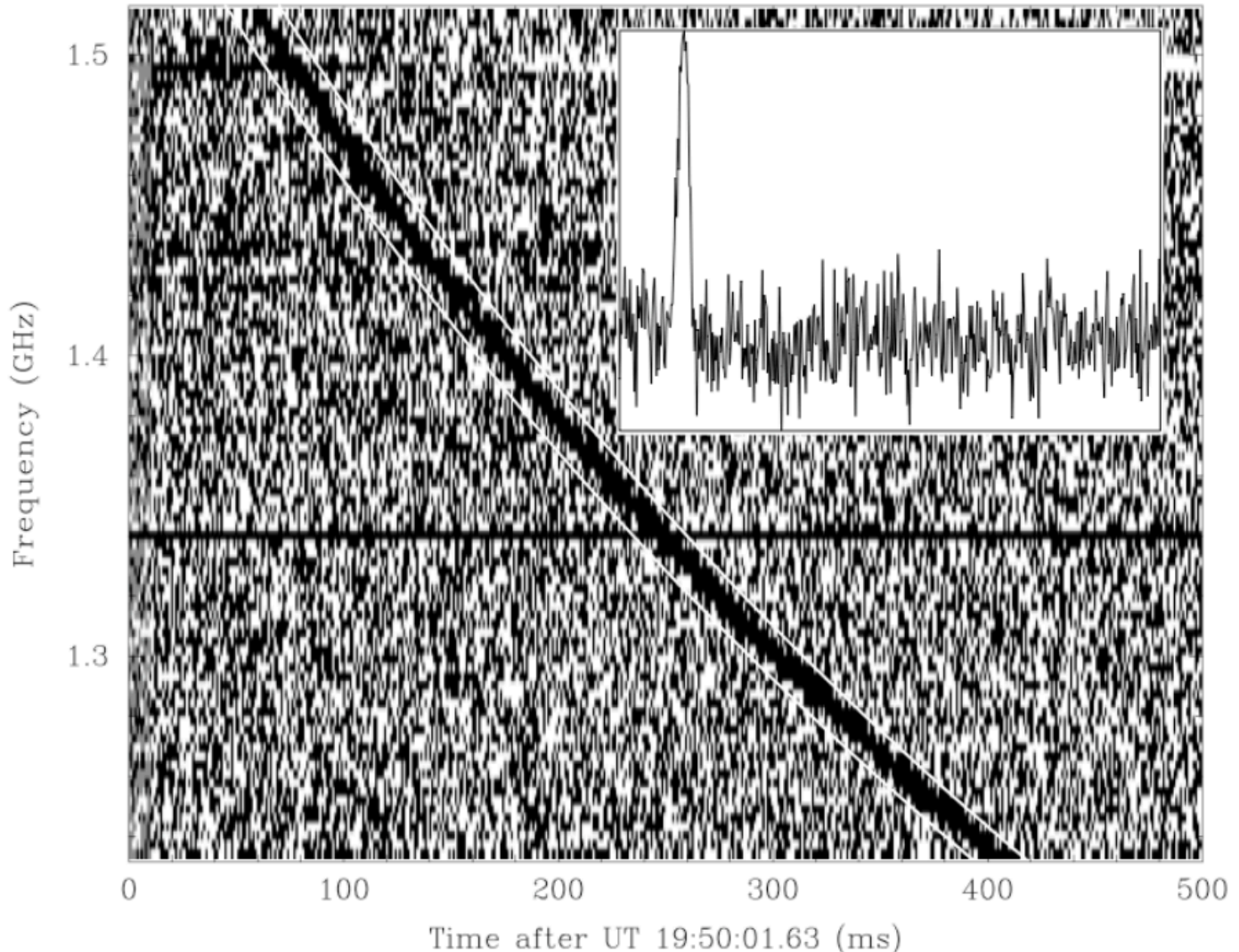
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Science Case: Fast Radio Bursts (FRBs)

- FRBs are rather frequent astrophysical phenomena lasting a few millisecond with a surprisingly large luminosity.
- They are observed by radio telescopes.
- The first FRB observation occurred in 2007, by Duncan Lorimer, astrophysicist at the West Virginia University, with the Parkes radiotelescopes.

LORIMER BURST (2001 data, published in Lorimer et al. 2007)



- The event was short and with a clear frequency drift. The dispersion measure was much higher than expected for a Galactic source, supporting an extra-galactic origin.

► Do you know what a distance measure is?

- It is interesting to see that in the same period radiotelescopes detected several more events similar to the Lorimer's burst called "peryttons".
- In spite of their similarity, it was inferred that perytons are the result of premature opening of a *microwave oven door*, which releases a frequency-swept radio pulse, which mimics a fast radio burst, as the magnetron turns off (from Wikipedia)!
- This generated a considerable confusion, at the time, until further research and observations clarified that FRBs were real astrophysical phenomena (Thornton et al. 2013).

- An important discovery came a few years later, by Keane et al. (2016), when the host galaxy of a FRB event was firmly identified.
 - The event, FRB150418, was localized in the galaxy at $z \sim 0.49$, confirming the cosmological nature of (at least some of) these events.
- Another important discovery occurred when FRB 200428, a nearby and faint FRB, was localized from a position consistent with an already known source: the magnetar SGR1935+2154.
 - It is still unclear to which extent this identification can play a role for the whole population of cosmological FRBs, but there is a large consensus that FRBs are indeed mainly cosmological although they can occur even locally.
- In several cases, as we are going to discuss later, FRBs have been observed to repeat, i.e. to produce multiple events from the same coordinates.
 - So the question. Are all the GRBs repeaters or are there two (at least) different populations?
- They also are often high polarised.
- In any case, the rate of FRBs is high: $> 1000\times$ the GRB rate.
- There are still several ideas about their origin:
 - They could be originated by neutron star (or maybe white dwarf too) collisions.
 - Else, they could be generated by phenomena related to pulsars, possibly with some sort of pseudo-periodicity.
 - Finally, again related to neutron stars, FRBs can be due to magnetars, highly magnetized neutron stars.
- In some more detail:

1. Magnetar Flares

- The most widely accepted theory is that FRBs are caused by magnetars—neutron stars with magnetic fields trillions of times stronger than Earth's.
- Emissions are supposed to be fed by “Starquakes” on the surface of the magnetar or explosive reconfigurations of its magnetic field release massive amounts of energy.
- As a matter of fact, the detection in 2020 of a burst from a magnetar within our own galaxy (SGR 1935+2154), is supposed (yet, see above) the “smoking gun” for this idea.

2. Colliding Compact Objects

- Some FRBs might be the “final result” of a cataclysmic collision, e.g., the merger of two neutron stars or a neutron star and a black hole.
- These events would be “one-offs.” While this could explain non-repeating FRBs, it cannot explain the repeater sources that flash multiple times from the same spot.

3. Collapsing Neutron Stars

- This phenomenon can occur when a heavy, spinning neutron star loses enough rotational speed that it can no longer support its own weight against gravity.
- As it collapses into a black hole, its magnetic field is suddenly snapped off or “shed,” producing a massive radio pulse.

4. Supermassive Black Hole Environments

- Some FRBs have been found in very extreme environments, such as the centers of galaxies near supermassive black holes.
- The interaction between a neutron star's plasma and the intense winds or gravity of a nearby supermassive black hole could amplify radio emissions to the levels we observe.

5. Interaction with Asteroids or Planets

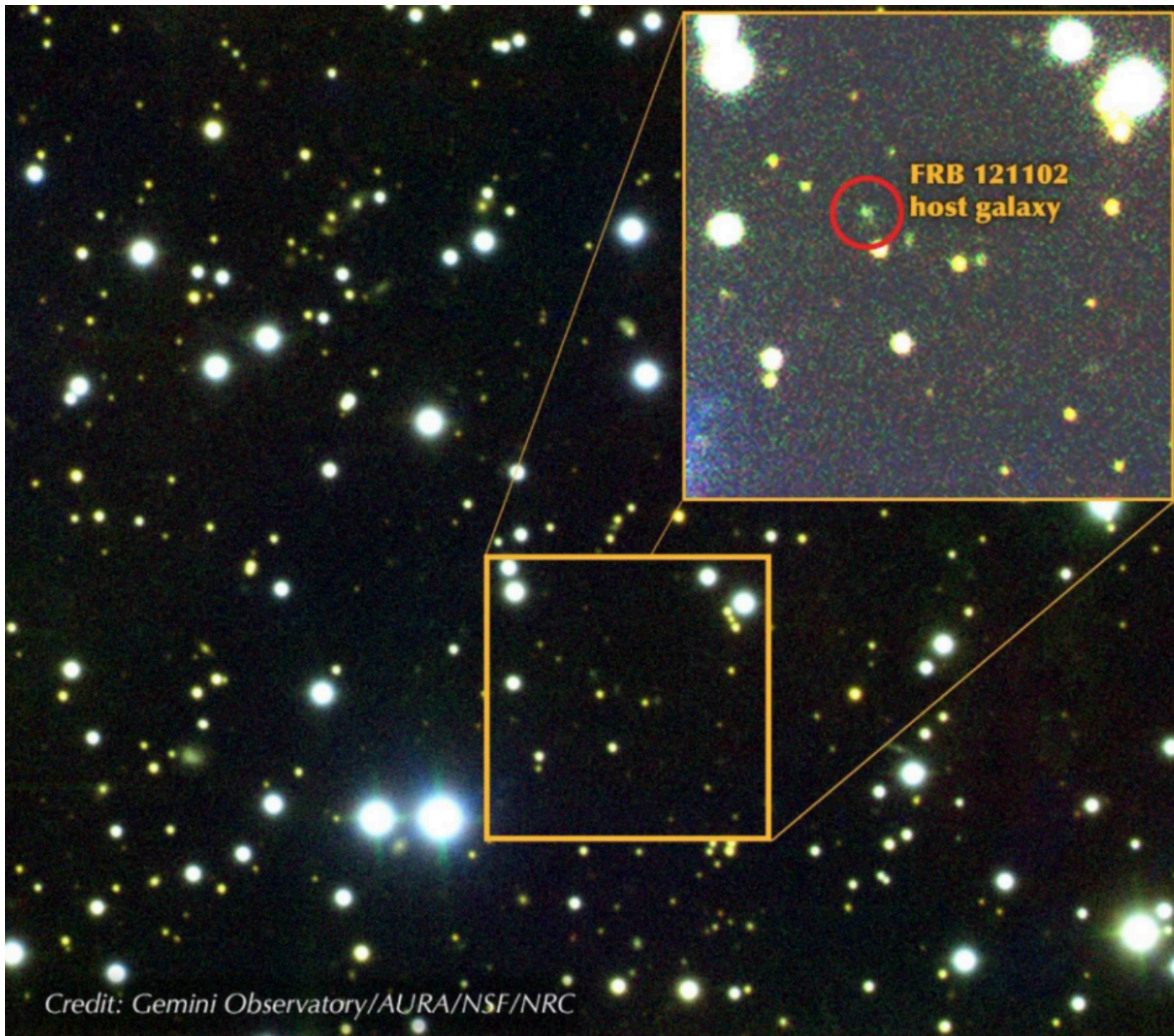
- A more local theory, i.e., within a star system, suggests that objects like asteroids or “hot Jupiters” might be orbiting a neutron star.
- As these objects pass through the intense stellar wind of the neutron star, they create a wake that generates powerful radio beams, similar to how Jupiter’s moon Io interacts with Jupiter’s magnetic field.

Repeating FRBs

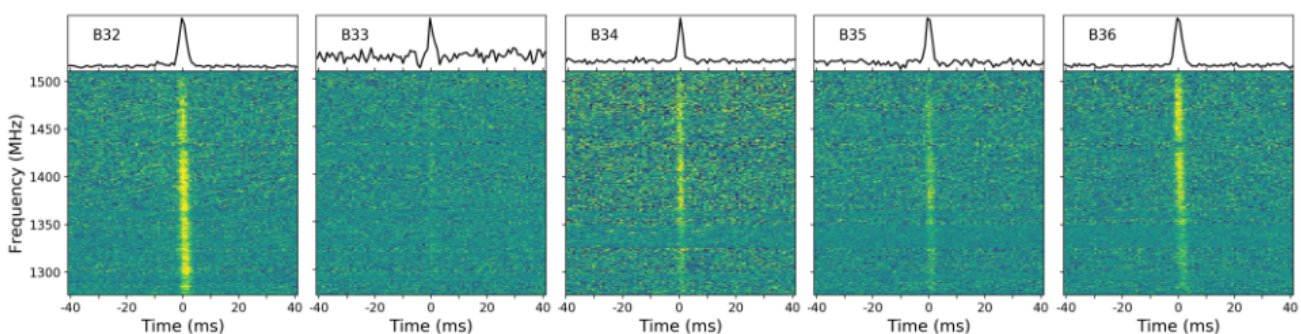
- Several FRBs (less than 10% of the total population, though) are observed to repeat with, possibly, some regularity. The topic is still highly debated, and of course can deeply affect the zoo of models proposed for these sources (e.g., [Wei et al. 2022](#)).
- FRB 180916.J10158+56 seems to show activity cycles with a period of 16.35 days ([Chime/Frb Collaboration 2021](#)), while FRB 121102 is active for about 90 days with a quiet period of about 67 days with a global "period" of roughly 157 days.
- These analyses are complex because it is needed to model the sensitivity of the telescopes that is not always uniform in time and might, by itself, introduce some distortion.
- The observing campaigns have been, unavoidably, non-uniform introducing, therefore, the whole set of problems related to uneven sampling and gaps.

FRB 121102

- We devote some attention to this event.



- After the claim for a possible periodicity intense observational campaigns have been devoted to this source.
- The waiting time between consecutive events can be an important diagnostics. For this FRB the distribution has been modelled as a Weibull distribution with $k < 1$ or as a regular Poisson distribution (see [Cruces et al. 2021](#)).
- The energy distribution of the FRB pulses with energies between $10^{38} - 10^{39}$ erg has been modeled as a power-law with slope $\gamma \sim -1.1$.



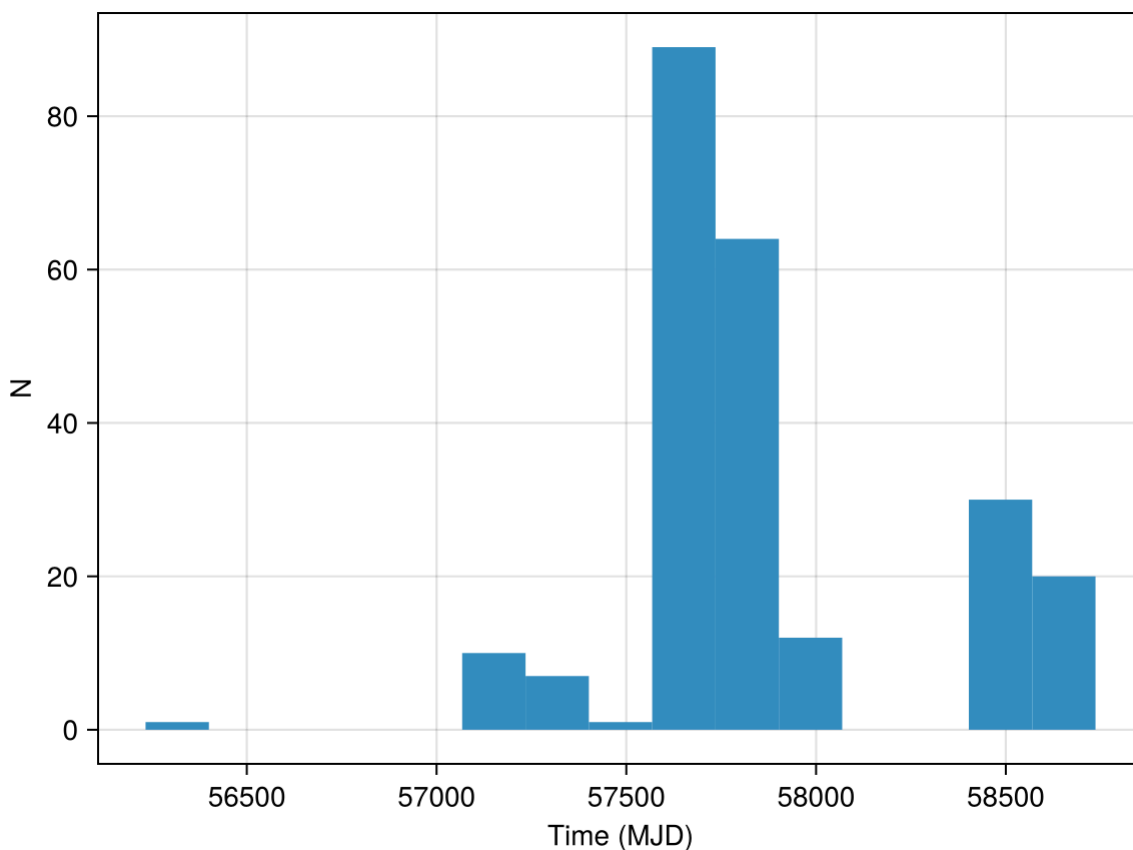
- Some of the pulsed detected for FRB 121102.

Exercise: Analyse the FRB121102 by the Gregory & Loredo algorithm

- We are going to the dataset provided by [Rajwade et al. \(2020\)](#):

```
1 begin
2   data = Vector{Float64}{}
3   open("All_dets_paper.txt") do f
4     l = readlines(f)
5     for i in l
6       push!(data, parse(Float64, split(i)[2]))
7     end
8   end
9 end
```

- We apply the [Gregory & Loredo \(1992\)](#) algorithm.
 - It computes the likelihood of a set of arrival times originating from a periodic system rather than constant rate (Poisson) background noise.
- Let's check how the arrival times are distributed:



Data consists of 234 events.

- The dataset covers about 2500 days, with most of the activities concentrated in a 500 day long interval.
- We need to define a frequency range for the analysis.
- We chose a logarithmic spacing.

```
1 begin
2     wr = FrequencyRange(data,logspacing=true)
3
4     whi = maximum(wr)
5     wlo = minimum(wr)
6
7 end;
```

Minimum period: **10.7** days, maximum period **250.3**, number of periods: **117**

Model parameters

- The Gregory & Loredó (1992) algorithm does not assume any specific light-curve shape, and describe the variability pattern as a box-shaped function.
- The maximum number of boxes is, in principle, only limited by the available computing power and data quality.
- For the dataset we are considering, we adopt a maximum number of boxes of 40.

```
1 mmax = 40;
```

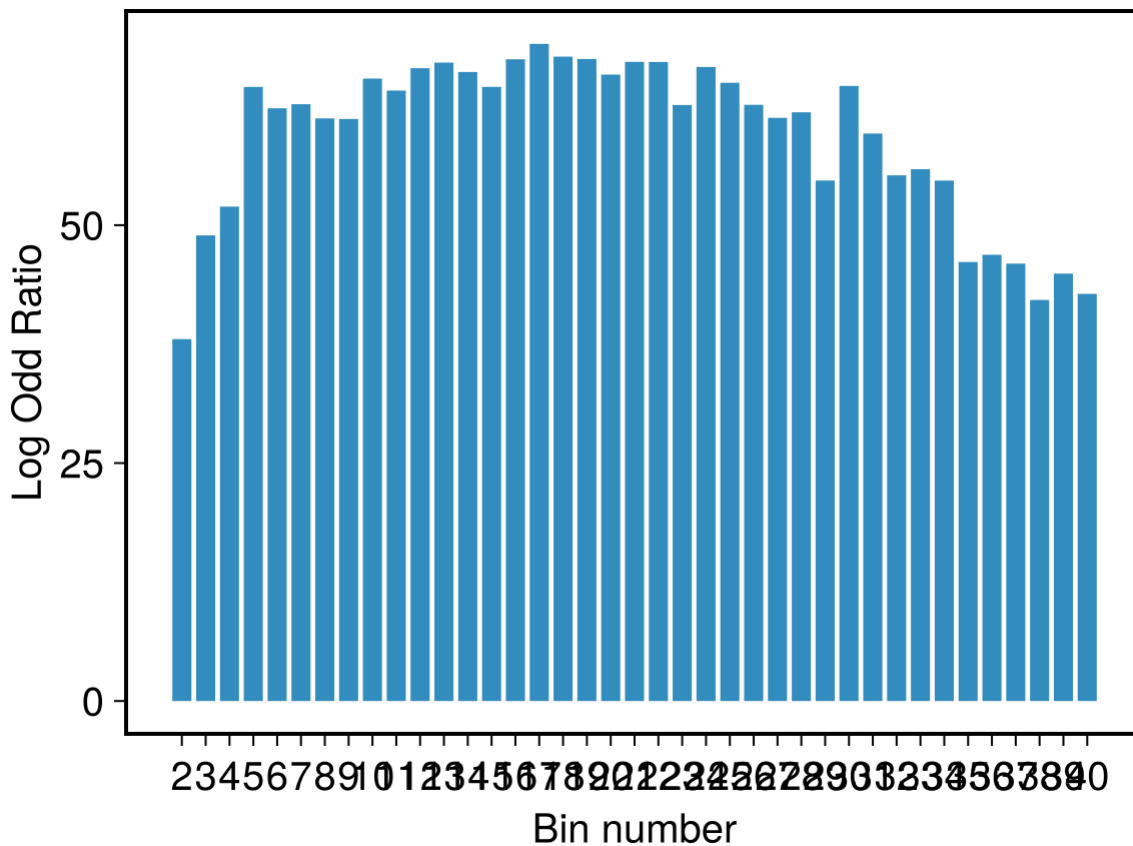
Odd ratios

- The first step is therefore to compute the odd ratios for each model (m from 2 to $m_{\max}$) compared to the constant (no variability model, $m=1$).

```
1 omq = OddRatios(data,wr,mmax);
```

- And plot the histogram of the obtained odd ratios:

Odd ratio log plot? ☒



- We can also derive some summary from the odd ratios distribution:

```
1 odrs = OddRatioSummary(omq);
```

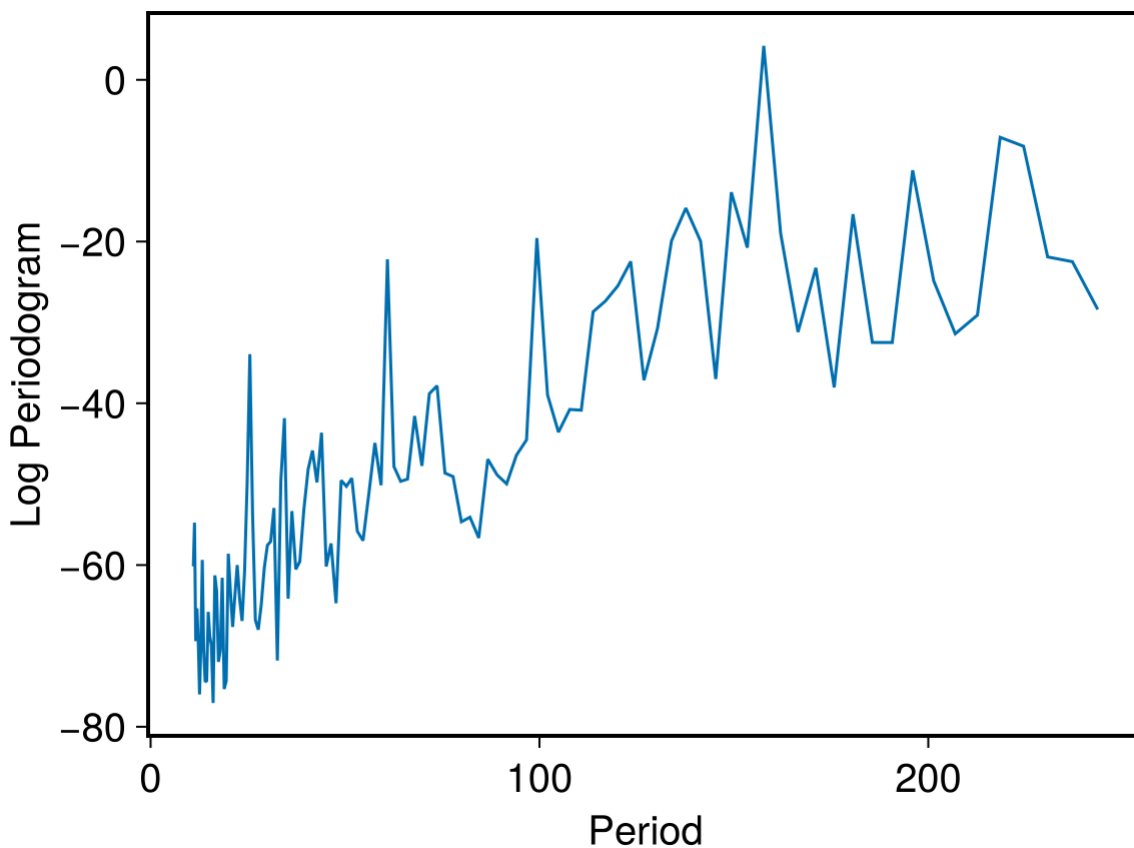
Probability non-constant model: **100.00%** and best number of bins: **17**

- First of all, we see that the probability for a non-constant model is extremely high, and apart from the simplest case with low m , the most probable models require m in the range approximately from 6 to 30.
- The highest probability is for $m=17$. Of course, pay attention to the logarithm scale of the plot.

Periodogram for the best m

- Let's therefore compute and plot a periodogram for this choice of m :

Periodogram log plot? 



- And let's derive some information from the periodogram data:

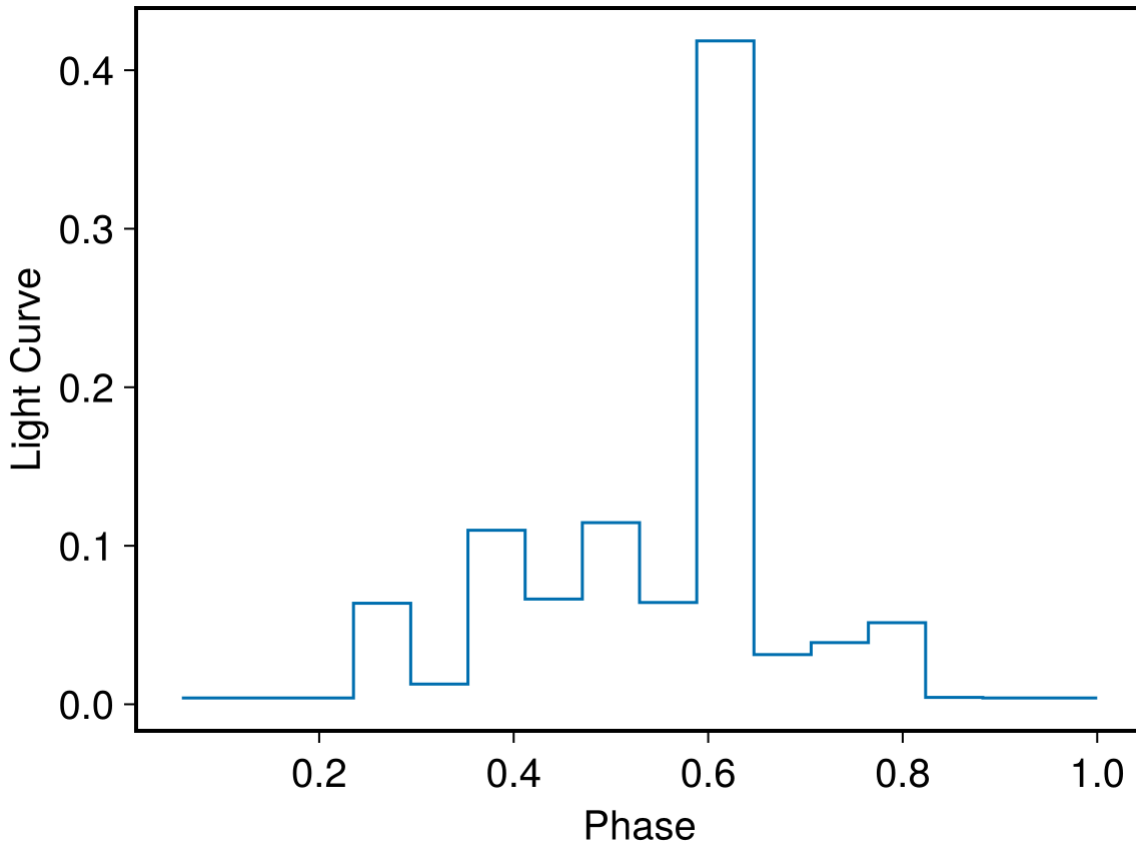
```
1 mpers = PeriodogramSummary(mper,wr);
```

Log maximum power: **4.20**, angular frequency at maximum: **0.040**, period: **157.69** days

- The periodogram indeed shows a rather prominent peak at the period = 157.69 days. This is in reasonable agreement with what reported in [Raywade et al.\(2020\)](#), although based on different tools.
- It is also of interest comparing our periodogram with what reported in their paper, in [Fig. 3](#).

Light-curve shape

- Having derived the most probable frequency for a periodic behaviour, we can compute and plot the light-curve shape:



```

1 begin
2   llpl = LightCurveShape(wr,odrs.maxm,data)
3   PlotLightCurve(llpl)
4 end

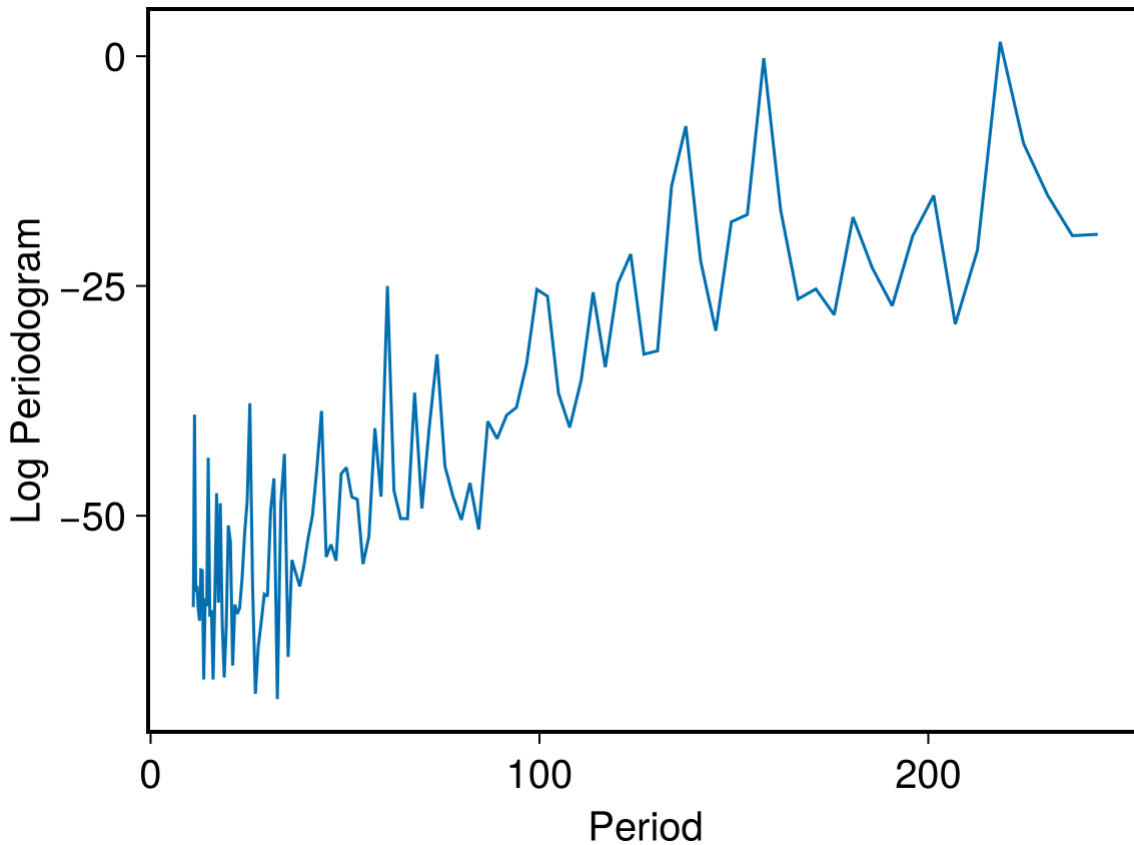
```

- And, again, we can compare it with the analogous plot in [Raywade et al.\(2020\)](#), i.e., their [Fig. 4](#).
- The two light-curves are very similar but now the phase is not *assumed*, it is just computed from the data, apecularity of the Gregory & Loredó (1992) algorithm.

Marginalized periodogram

- Finally, rather than choosing the most probable m , it is also possible, and probably better justified, computing a periodogram marginalizing for m .
- Results are often very similar, specially if the most probable m stands clearly out the general distribution. But this is not always the case.
- The marginalized periodogram can be computed and plotted with:

Periodogram log plot? ☒



- And let's see the usual summary:

```
1 marmpers = PeriodogramSummary(marmper, wr);
```

Log maximum power: **1.56**, angular frequency at maximum: **0.029**, period: **218.49** days

- As a matter of fact, the peak at ~157 days is still there, but a new peak at ~218 days appears.
- Both peaks are now less prominent, possibly suggesting that the identified periodicity is not (basing upon the available data) very solid.

Reference & Material

Material and papers related to the topics discussed in this lecture.

- [Rajwade et al. \(2020\) - "Possible periodic activity in the repeating FRB 121102"](#)
- [Cruces et al. \(2021\) - "Repeating behaviour of FRB 121102: periodicity, waiting times, and energy distribution"](#)

Further Material

Papers for examining more closely some of the discussed topics.

- [Thornton et al. \(2013\)](#) - "A Population of Fast Radio Bursts at Cosmological Distances"
- [Keane et al. \(2016\)](#) - "The host galaxy of a fast radio burst"
- [Chime/Frb Collaboration \(2021\)](#) - "Periodic activity from a fast radio burst source"
- [Wei et al. \(2022\)](#) - "The periodic origin of fast radio bursts"

Course Flow

	Previous lecture	Next lecture	Course Summary
notebook	Lecture about time of arrival analysis	Lecture about non-parametric analysis	Course Summary
html	Lecture about time of arrival analysis	Lecture about non-parametric analysis	Course Summary

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